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Circulating renin–angiotensin–aldosterone system markers in cats with non-hypertensive chronic kidney disease or systemic arterial hypertension

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Abstract

Background

Information on the classical and alternative circulating renin–angiotensin–aldosterone systems (RAAS) in cats with non-hypertensive chronic kidney disease (NHT-CKD) or systemic arterial hypertension (SAH) is limited. Age and diet affect the RAAS.

Hypothesis/Objectives

To compare serum equilibrium concentrations of angiotensin peptides and aldosterone in healthy cats and cats with NHT-CKD or untreated SAH, and to evaluate changes in these concentrations in hypertensive cats administered amlodipine.

Animals

Client-owned cats with NHT-CKD (serum creatinine ≥ 1.6 mg/dL, systolic blood pressure [SBP] < 160 mmHg; $n = 17$), SAH (SBP ≥ 160 mmHg or ≥ 150 mmHg with hypertensive ocular lesions; $n = 6$), or normal SBP (< 140 mmHg) and kidney function ($n = 17$).

Methods

Prospective, single-center, observational study. Cats underwent indirect SBP measurement and clinicopathologic analyses. Serum was obtained contemporaneously in all cats, and after 2-4 weeks of amlodipine therapy in hypertensive cats, for evaluation of angiotensin I, II, III, IV, 1-5, and 1-7, and aldosterone.

Results

Adjusting for diet, age, or both modified the conclusions of statistical comparisons for several variables. Mean serum angiotensin I, II, and III concentrations were lower in cats with untreated SAH vs healthy controls across unadjusted and diet-, age-, or both diet- and age-adjusted linear mixed models (geometric mean [95% CI] angiotensin I, 9.02 [3.05-26.62] vs 29.78 [21.15-41.94] pmol/L; angiotensin II, 33.63 [12.25-92.26] vs 124.24 [90.52-170.51] pmol/L; angiotensin III, 1.56 [0.88-2.79] vs 7.96 [5.35-11.86] pmol/L; all $P \leq .038$). Controlling for diet but not age, angiotensin II ($P = .043$) and III ($P = .019$) were also lower in cats with NHT-CKD vs controls. All angiotensin peptides, except for angiotensin IV, significantly increased in hypertensive cats during amlodipine therapy. Serum aldosterone did not differ between groups, or before and after amlodipine in hypertensive cats.

Conclusions and clinical importance

Evidence of circulating RAAS activation in cats with NHT-CKD or untreated SAH was not found.

Keywords: alternative RAAS, amlodipine, classical RAAS, healthy cats

Introduction

The renin–angiotensin–aldosterone system (RAAS) plays a pivotal role in the regulation of kidney function and the pathophysiology of kidney diseases.¹ Although the RAAS is commonly said to be activated in chronic kidney disease (CKD),² this is inconsistently supported by published studies of its circulating markers in cats with kidney dysfunction.³ Plasma renin activity (PRA) and angiotensin II and aldosterone concentrations were higher in cats with experimentally induced CKD relative to preinjury values in 1 study,⁴ as well as in cats with spontaneous CKD and unknown blood pressure (BP) status relative to healthy cats in another.⁵ In other studies, cats with naturally occurring hypertensive CKD had unchanged or lower PRA and higher plasma aldosterone concentrations relative to cats without systemic arterial hypertension (SAH).⁶⁻⁸ Overall, patterns of circulating RAAS (cRAAS) activity appear dependent on certain disease characteristics, such as whether azotemia or SAH is present,^{8,9} and concurrent treatment with amlodipine, an L-type calcium channel blocker known to activate the RAAS.⁸ Age, sex, and dietary sodium and potassium intake also influence RAAS activity.¹⁰⁻¹²

The RAAS is composed of 2 pathways that mediate opposing effects. The first is the so-called *classical* pathway, which culminates in the binding of angiotensin II to its type 1 receptor and mediates vasoconstriction and renal sodium and water retention, among other physiologically relevant effects.¹³ The second is the counterbalancing *alternative* pathway, which culminates in the interaction of angiotensin II with its type 2 receptor and the interaction of angiotensin 1-7 peptide (primarily generated by angiotensin-converting enzyme 2 [ACE2]) with the Mas receptor, both of which mediate vasodilation, natriuresis, and diuresis.¹⁴ To date, most studies of the RAAS in cats with CKD have focused on a limited number of cRAAS activity markers, such as PRA, angiotensin II, and aldosterone.^{2,5-9,15} Two studies describe the activities of both the classical and alternative pathways in cats with SAH, some of which had CKD.^{16,17} However, extensive characterization of the cRAAS in cats with non-hypertensive, naturally occurring CKD is lacking.

The objectives of the present study were (1) to describe equilibrium concentrations of circulating angiotensin peptides and aldosterone in cats with naturally occurring International Renal Interest Society (IRIS) stages 2-4, non-hypertensive chronic kidney disease (NHT-CKD), in cats with naturally occurring SAH, and in apparently healthy cats; and (2) to characterize changes in the equilibrium concentrations of circulating angiotensin peptides and aldosterone in hypertensive cats started on amlodipine, relative to pretreatment values. We hypothesized that (1) compared with age-matched, healthy control cats, cats with NHT-CKD or SAH would have angiotensin peptide and aldosterone concentrations and estimated renin activity levels that indicate activation (NHT-CKD and SAH groups) or a lack of appropriate suppression (SAH group) of the cRAAS; and that (2) antihypertensive treatment with amlodipine besylate would significantly increase angiotensin peptide and aldosterone concentrations.

Materials and methods

Study design

This was a prospective, cross-sectional, single-center, observational study conducted between March 2021 and September 2023. In all cats, the cRAAS was evaluated by equilibrium analysis at a single time point. For hypertensive cats prescribed amlodipine by the attending clinician, cRAAS markers were also reevaluated 2-4 weeks after initiating therapy. All study activities were approved by the Clinical Research Committee of the University of Georgia College of Veterinary Medicine (approval number CR-594). Informed owner consent was obtained from all cat owners prior to enrollment.

Animals

Eligible participants were companion cats of any age, sex, or breed with IRIS stages 2-4 NHT-CKD or SAH of any etiology, presented routinely to the University of Georgia Veterinary Teaching Hospital or recruited from staff- and student-owned cats. In addition, healthy control cats that were age- and sex-matched to NHT-CKD cats were recruited from staff- and student-owned cats during the same period.

Screening visit

For all cats, a screening visit included a complete medical history; a physical examination including indirect fundic evaluation; systolic blood pressure (SBP) measurement using Doppler sphygmomanometry and guidelines set forth by the American College of Veterinary Internal Medicine¹⁸; a CBC, a serum biochemistry profile including symmetric dimethyl arginine (SDMA) concentration, and a total thyroxine (TT4) concentration; urinalysis and a urinary protein-to-creatinine ratio (UPC); and transthoracic echocardiographic evaluation (for all but one NHT-CKD cat). The one cat not undergoing echocardiography was the first enrolled in the study and had no historical or physical examination abnormalities suggestive of cardiovascular disease. Anxiolytics or sedatives were not administered to any cat before SBP determination.

Enrollment criteria

Cats were excluded if they had an unsuitable temperament; had concurrent disease(s) or condition(s) with a known or likely effect on the RAAS (eg, heart disease with any degree of left

atrial enlargement, congestive heart failure, volume depletion, diabetes mellitus, and uncontrolled hyperthyroidism); or had received a RAAS inhibitor (eg, angiotensin-converting enzyme inhibitor, angiotensin receptor blocker, and mineralocorticoid receptor blocker) in the 30 days preceding enrollment, had received a corticosteroid by any administration route in the 30 days preceding enrollment or in the 6 months prior to enrollment for depot formulations; had received other medications known to modify the RAAS directly or indirectly (eg, amlodipine, hydralazine, β - or α -adrenergic blockers, and diuretics) in the 7 days preceding enrollment; or if the owner did not consent to participation.

Control group

Cats were included in the healthy control group based on normal findings on physical examination; indirect SBP measurement (normal, < 140 mmHg); CBC; serum biochemistry profile, SDMA, and TT4; urinalysis; UPC; and echocardiography. Cats were deemed to have normal urine concentrating ability if the urine specific gravity (USG) was ≥ 1.035 . Cats with a spot USG of 1.030-1.034 during screening were also eligible for enrollment provided physical examination and clinicopathologic variables reflective of water balance suggested optimal hydration status, owners reported normal drinking and urination, and if subsequent renal ultrasonographic examination (performed for other unrelated research studies in some cats) was normal, USG at follow-up testing was ≥ 1.035 , or both. To be screened for inclusion as a healthy control, cats had to be age-, sex-, and neuter-status-matched as a matched-pair with a cat enrolled in the NHT-CKD group. Control cats were considered age-matched if the age difference was $\leq 20\%$ (using the NHT-CKD cat as a reference).

NHT-CKD group

Cats were included in the NHT-CKD group if diagnosed with CKD based on evidence of chronic changes in renal function and assigned an IRIS CKD stage of 2 or higher (ie, serum creatinine concentration ≥ 1.6 mg/dL with contemporaneous USG < 1.035 and the absence of an apparent extra-renal cause) and substaged as normotensive (ie, SBP < 140 mmHg) or prehypertensive (ie, SBP, 140-159 mmHg). Cats of any proteinuria substage were included. Systemically non-hypertensive cats with concurrent hyperthyroidism (controlled or uncontrolled) were excluded from the NHT-CKD group.

Hypertensive group

Cats were included in the SAH group if an average session SBP ≥ 160 mmHg was documented during 2 separate measurement sessions at least 1 h apart, or if a single average session SBP ≥ 150 mmHg was noted in a cat with ocular target organ damage (eg, retinal hemorrhage, bullous retinal detachment). Cats could be diagnosed with idiopathic or secondary hypertension. Historically hyperthyroid cats definitively treated with I-131 and whose hyperthyroidism was controlled (ie, serum TT4 concentration $\leq$ upper limit of the laboratory reference interval) for at least 4 weeks prior to screening could be included. All treatment decisions were at the discretion of the attending clinician. Hypertensive cats prescribed amlodipine alone at any dosage were invited back 2-4 weeks after initiation of therapy for repeat evaluation.

Follow-up visit for amlodipine-treated cats

For hypertensive cats prescribed amlodipine at the screening visit, a follow-up visit included a physical examination with indirect fundic evaluation; indirect SBP measurement; and blood and urine sampling for a renal biochemistry profile, urinalysis, UPC measurement, and cRAAS evaluation.

Sample collection and clinicopathologic analysis

At all visits, after SBP determination and physical examination, blood and urine samples were collected via venipuncture and cystocentesis, respectively. Samples were most often collected in unsedated cats. For 8 cats ($n = 5$ and 3 in the NHT-CKH and healthy control groups, respectively), butorphanol (0.2-0.3 mg/kg IM or IV) was administered to facilitate sample collection. Blood was collected from the jugular, saphenous, or cephalic vein and immediately placed into tubes containing no additive, where it was allowed to clot, or EDTA. Urine was collected into glass tubes. Hematology, serum biochemistry, serum TT4 concentration, urinalysis, and UPC determination were performed in-house by the University of Georgia Veterinary Clinical Pathology Laboratory (Athens, Georgia). Serum was shipped on ice to a commercial laboratory (IDEXX Laboratories, Westbrook, Maine) for measurement of SDMA concentration. Within 30 min of collection, serum was separated by centrifugation and stored at -80°C until batch shipment for measurement of cRAAS analytes as described below.

Cats were fasted overnight and food was withheld on the morning of evaluation. Although time of day of sample collection was not standardized, most visits occurred between 9 AM and 12 PM; when possible, reevaluation of hypertensive cats happened at the same time of day as the corresponding screening visit.

Analysis of cRAAS components

The cRAAS was assessed by measurement of serum equilibrium concentrations of angiotensin I, II, III, IV, 1-5, and 1-7 and aldosterone using a previously described LC–MS/MS-based technique (RAS-Fingerprint; Attoquant Diagnostics GmbH, Vienna, Austria).¹⁹ In addition, surrogate markers of enzyme activity and adrenal responsiveness to angiotensin II signaling were calculated from these concentrations, as previously described.²⁰⁻²² These included surrogate measures of ACE activity ($\text{ACE-S} = [\text{angiotensin II}] / [\text{angiotensin I}]$; ACE2 activity ($\text{ACE2-S} = [\text{angiotensin 1-5}] / [\text{angiotensin II}]$; PRA ($\text{PRA-S} = [\text{angiotensin I}] + [\text{angiotensin II}]$); and adrenal responsiveness to angiotensin II (aldosterone-to-angiotensin II ratio, AA2 ratio) = $[\text{aldosterone}] / [\text{angiotensin II}]$.

Echocardiography

All echocardiographic examinations were performed by a board-certified veterinary cardiologist or cardiologist-in-training, using an ultrasound unit (EPIQ CVx, Philips Healthcare) outfitted with phased-array transducers. Complete transthoracic echocardiographic studies were recorded from standard right and left parasternal views on manually restrained animals.²³ Findings were used to guide enrollment decisions as described above. Diagnosis and classification of cardiomyopathies conformed to guidelines set forth by the American College of Veterinary Internal Medicine.²⁴ The hypertrophic cardiomyopathy (HCM) phenotype was assigned to cats with left ventricular concentric hypertrophy (ie, end-diastolic wall thickness ≥ 6 mm) and a non-dilated left ventricle.²⁴ Left atrial enlargement was diagnosed in cats with a left-atrial-to-aortic root diameter ratio (LA:Ao, see below) > 1.5 .^{25,26}

For the purposes of statistical analyses, the following data were recorded for each enrolled cat: (1) LA:Ao calculated from 2-dimensional (2D) short-axis measurements,²⁶ (2) maximum 2D- or M-mode-derived end-diastolic left ventricular wall thickness at any segment (maxLVWTd, mm; a measure of left ventricular hypertrophy severity), and (3) left ventricular fractional shortening (FS, %; a measure of ventricular systolic performance) calculated from M-mode measurements. For each variable, the recorded value reflected the average of values taken from 3 to 5 cardiac cycles.

Statistical analyses

Statistical analyses were performed using commercially available (Graphpad Prism 10 for MacOS, version 10.4.2) and open-source (R project, version 4.4.1²⁷) software packages. For all analyses, P -values $< .05$ were considered statistically significant.

Continuous data were evaluated for normality of distribution by visualization of histograms and the Shapiro–Wilk test. Baseline demographic and clinical data are presented as mean $\pm$ SD or median (first quartile; third quartile) if normally or non-normally distributed, respectively. For RAAS data compared with linear mixed models, which were log-transformed as described below, geometric means and their 95% CIs were also calculated. For the purposes of statistical analysis, USG values > 1.060 were assigned a value of 1.061. Angiotensin peptide concentrations less than the assay lower limit of quantification were reported as that limit divided by 2.

For normally and non-normally distributed variables, intergroup comparisons were performed using a one-way analysis of variance (ANOVA) test or the Kruskal–Wallis test, respectively, with pairwise comparisons adjusted for using the Benjamini–Hochberg method. Categorical data were compared using Fisher’s exact test.

Linear mixed models were built using the R package lme4²⁸ (version 1.1-37), and marginal means and group comparisons were calculated with the package emmeans²⁹ (version 1.11.0). Default package settings were used unless otherwise specified. Angiotensin peptide and aldosterone concentrations, and RAAS surrogate indices calculated from these concentrations, were log-transformed to improve the approximation of a normal distribution. Linear mixed models were then used to compare RAAS markers between the 3 groups. The group (ie, healthy control, NHT-CKD, or SAH) was set as the fixed effect with the healthy control group set as the reference category. Pairing between healthy control and NHT-CKD cats was set as a random effect and the RAAS marker was set as the dependent variable. Multiple comparisons between the 3 groups were adjusted for using the Tukey method. Additional models were built with the feeding of a therapeutic renal diet, age, or both, also set as a fixed effect to adjust for possible effects of diet or age on these group comparisons. In SAH cats only, the Wilcoxon signed rank test was used to evaluate differences in RAAS markers before vs after amlodipine treatment.

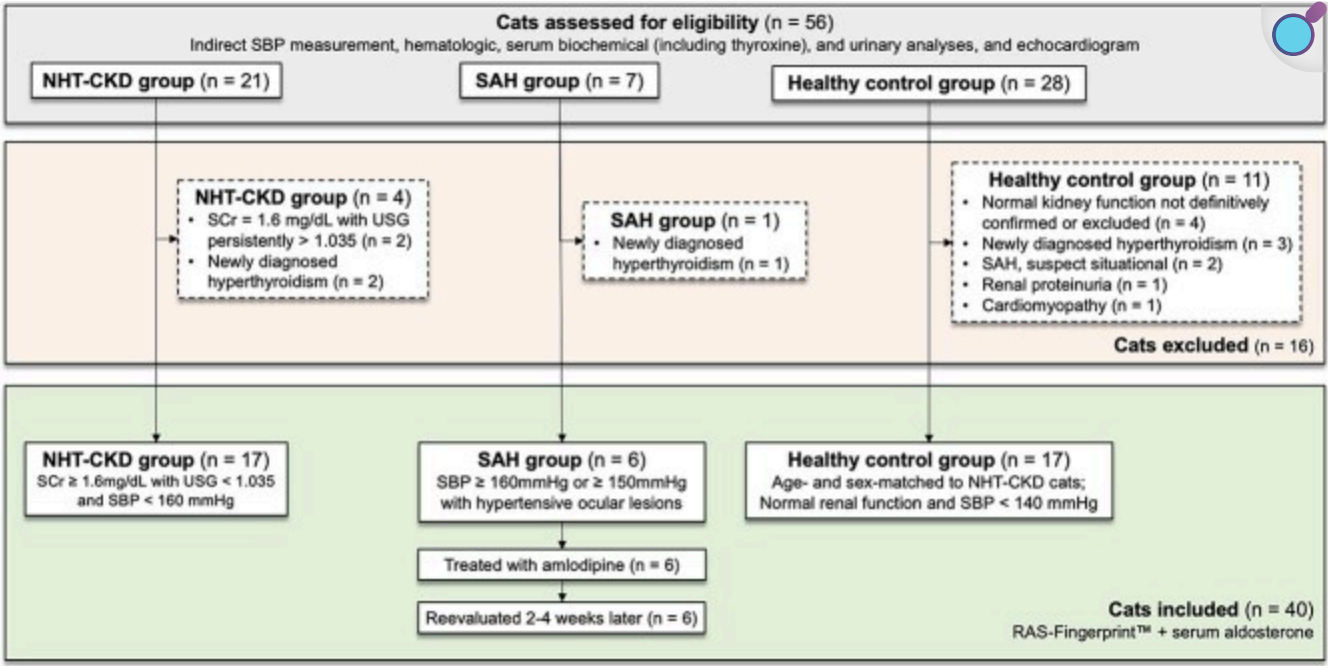
Because of the known interaction between the RAAS and electrolyte balance, and kidney and cardiovascular function,³⁰ exploratory Spearman correlation analysis was conducted between the cRAAS markers evaluated and selected clinical, clinicopathologic, and echocardiographic variables. Specifically, correlations between RAAS markers and SBP; hematocrit; serum sodium, potassium, creatinine, and SDMA concentrations; UPC; and 3 echocardiographic variables (ie, LA:Ao, maxLVWTD, and FS) were explored. Data from all enrolled cats were used in correlation analysis with only pre-amlodipine values used for SAH cats.

Results

Animals

Fifty-six cats were screened for eligibility, of which 40 were included ($n = 17, 17$, and 6 in the NHT-CKH, healthy control, and SAH groups, respectively; [Figure 1](#)). The most common reasons for exclusion included newly diagnosed hyperthyroidism ($n = 6$ cats) and inability to definitively establish normal kidney function in cats screened for the healthy control group ($n = 4$ cats).

Figure 1.



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Flow diagram illustrating the progress of cats through the present study, which evaluates the circulating renin–angiotensin–aldosterone system in cats with NHT-CKD in International Renal Interest Society stages 2 or higher, cats with SAH (all of which had CKD), and healthy controls. Abbreviations: NHT-CKD = non-hypertensive chronic kidney disease; SAH, systemic arterial hypertension.

For cats included in the SAH group, SBP ranged from 157 to 182 mmHg. Evidence of hypertensive ocular target organ damage was noted in 4/6 (67%) cats, including the cat with the lowest SBP. Both treatment-naïve hypertensive cats without ocular lesions had a preexisting diagnosis of CKD.

In one of these cats without ocular lesions (with an average SBP at enrollment of 179 mmHg measured twice, 3 h and 15 min apart), mild aortic root dilatation and insufficiency were observed echocardiographically. In the other cat (with an average SBP at enrollment of 182 and 176 mmHg, measured 2 h and 20 min apart), SAH (SBP > 160 mmHg) was confirmed prior to enrollment using Doppler sphygmomanometry performed by the owner (a trained veterinary technician) in the home environment.

On average, cats in the NHT-CKD and SAH groups were moderately azotemic ([Table 1](#)). All 17 NHT-CKD cats were IRIS stage 2 ($n = 13$) or 3 ($n = 4$), with the majority ($n = 13$ [76%]) classified as normotensive and the remainder ($n = 4$ [24%]) classified as prehypertensive. Of these NHT-CKD cats, 11 (65%) were non-proteinuric, 3 (18%) were borderline proteinuric, and 3 (18%) were overtly proteinuric. All 6 cats in the SAH group had secondary hypertension and CKD of IRIS stage 2 ($n = 5$) or 3 ($n = 1$), with 4 (67%) and 2 (33%) being borderline or non-proteinuric, respectively. One hypertensive cat had a history of hyperthyroidism that was definitively treated with I-131 63 days prior to enrollment.

Table 1.

Demographic, clinical, and clinicopathologic information for enrolled cats with NHT-CKD (International Renal Interest Society stages 2 or higher), treatment-naïve SAH (all of which had CKD), or normal kidney function and blood pressure (healthy control group). Cats in the healthy control group were age- and sex-matched to those in the NHT-CKD group. Significant *P*-values are indicated in bold.

Variable	Healthy control group (<i>n</i> = 17)	NHT-CKD group (<i>n</i> = 17)	SAH group (<i>n</i> = 6)	<i>P</i> -value
Age (years)	11.6 (9.8; 14.1)	13.5 (12.2; 14.6)	15.9 (13.9; 17.3)	N/A ^{a,b}
Sex/neuter status (<i>n</i>)				N/A ^{a,b}
Female spayed	10	10	4	
Male neutered	7	7	2	
Body weight (kg)	4.0 (3.5; 5.2)	3.9 (3.6; 5.0)	4.2 (3.8; 4.8)	.99
Body condition score	5 (5; 7)	6 (5; 6)	6 (5; 7)	.92
Breed				
DSH	10	12	5	.57
DMH	3	0	0	
DLH	2	4	1	
Bengal	1	0	0	
Ragdoll	1	0	0	
Siamese	0	1	0	
Average SBP ^c (mmHg)	119 ± 14	128 ± 15	174 ± 10	N/A ^a
Hematocrit (%) (RI, 33.6-49.0)	40 ± 5 ^d	34 ± 4 ^d	33 ± 4 ^d	.002
Total protein (g/dL) (RI, 6.5-8.2)	7.7 ± 0.7	7.6 ± 0.6	7.7 ± 0.4	.94

Variable	Healthy control group (<i>n</i> = 17)	NHT-CKD group (<i>n</i> = 17)	SAH group (<i>n</i> = 6)	<i>P</i> -value
Serum creatinine (mg/dL) (RI, 0.7-1.8)	1.1 (1.0; 1.3)	2.1 (1.9; 2.8)	2.4 (1.9; 2.7)	N/A _{a,b}
Serum urea nitrogen (mg/dL) (RI, 18-35)	23 ± 3	44 ± 12	39 ± 7	N/A _{a,b}
Serum SDMA (µg/dL) (RI, 0-14)	10 ± 2	19 ± 6	20 ± 3	N/A _{a,b}
Serum sodium (mmol/L) (RI, 148-156)	151 ± 2	152 ± 2	152 ± 2	.32
Serum potassium (mmol/L) (RI, 3.7-5.3)	4.1 (4.0; 4.4)	4.3 (4.2; 4.5)	4.3 (4.1; 4.5)	.62
Serum total calcium (mg/dL) (RI, 9.5-11.0)	9.6 ± 0.5 _d	10.4 ± 0.6 _d	10.2 ± 0.2 _d	<.001
Serum phosphorus (mg/dL) (RI, 2.8-5.7)	3.5 (3.2; 4.3)	4.1 (3.9; 5.2)	4.1 (3.6; 4.5)	.08
Serum total thyroxine (µg/dL) (RI, 1.3-3.5)	2.5 ± 0.5	2.2 ± 0.6	2.0 ± 0.8	.1
Urine specific gravity	1.050 (1.047; 1.057)	1.019 (1.018; 1.024)	1.016 (1.015; 1.023)	N/A _{a,b}
Urinary protein-to-creatinine ratio	0.13 (0.11; 0.16)	0.18 (0.13; 0.28)	0.24 (0.11; 0.32)	N/A _{a,b}
LA:Ao	1.15 (1.10; 1.25)	1.25 (1.12; 1.31)	1.10 (1.02; 1.16)	.22
maxLVWTd (mm)	3.9 (3.3; 4.5) _d	4.5 (4.1; 5.3) _d	5.0 (4.7; 5.0) _d	.026
LVFS (%)	56 (54; 59)	57 (52; 59)	59 (57; 62)	.45

Data are shown as mean $\pm$ SD and compared with the one-way ANOVA test or median (first quartile; third quartile) compared with the Kruskal–Wallis test, as appropriate.

^aData not statistically compared between the 3 groups because it was part of the defining criteria for each group (eg, markers of renal function, systolic blood pressure) or it was used for case-matching (ie, age and sex for the NHT-CKD and SAH groups).

^bParameters did not differ significantly between the NHT-CKD and SAH groups using the Fisher's exact or Mann–Whitney *U* test, as appropriate.

^cFor each cat, 5 individual measurements were averaged to determine SBP.

^dGroups sharing a superscript letter are *not* significantly different at a 5% significance level. Pairwise comparisons were adjusted using the Benjamini–Hochberg method.

Abbreviations: ANOVA, analysis of variance; DLH, domestic longhaired; DMH, domestic medium hair; DSH, domestic shorthair; LA:Ao, left-atrial-to-aortic dimension ratio; LVFS, left ventricular fractional shortening; maxLVWTd, maximum end-diastolic left ventricular wall thickness; NHT-CKD, non-hypertensive chronic kidney disease; RI, reference interval; SAH, systemic arterial hypertension.

Echocardiographic findings were normal in 69%, 67%, and 100% of cats in the NHT-CKD, SAH, and healthy control groups, respectively. An HCM phenotype was diagnosed in 4 (25%) of 16 cats in the NHT-CKD group (the first enrolled cat did not have an echocardiogram), 3 of which also had systolic anterior motion of the mitral valve. Although no cat in the SAH or healthy groups had an HCM phenotype, the median maxLVWTd was significantly higher in SAH cats relative to healthy controls ([Table 1](#)). Discrete upper septal thickening³¹ was present in 1 SAH cat. In keeping with the study's enrollment criteria, left atrial enlargement was not present in any cat.

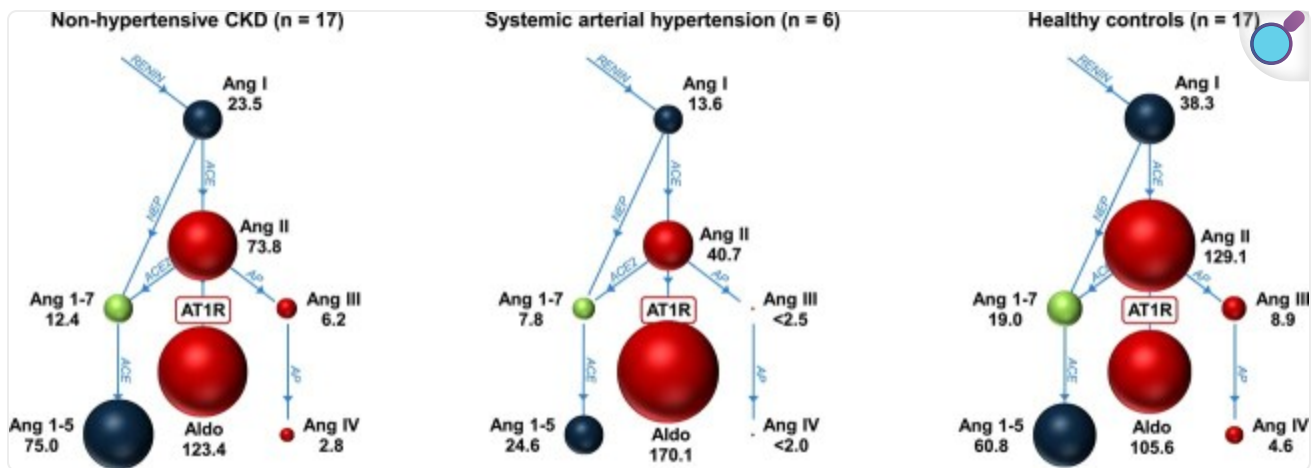
Cats were eating a variety of commercial diets, with a therapeutic renal diet being exclusively fed to 8 (47%) of 17 and 4 (67%) of 6 cats in the NHT-CKD and SAH groups, respectively. A complete list of fed diets is included in [Table S1](#).

Approximately one-half of the 40 included cats ($n = 9$ NHT-CKD, $n = 1$ SAH, and $n = 9$ healthy cats) were receiving no medications or supplements. Ectoparasite or heartworm preventives were being regularly administered to 35% of cats ($n = 4$ NHT-CKD, $n = 2$ SAH, and $n = 8$ healthy). Other medications being regularly administered included mirtazapine ($n = 1$ NHT-CKD cat), maropitant ($n = 1$ NHT-CKD cat), and SQ immunotherapy for atopy ($n = 1$ SAH cat). A list of administered supplements is included in [Table S2](#).

Comparison of cRAAS markers between groups

Median serum equilibrium concentrations of angiotensin peptides and aldosterone by group are depicted in [Figure 2](#). Geometric means and corresponding 95% CIs for these concentrations and calculated surrogates of RAAS activity are listed in [Table S3](#). Circulating RAAS markers did not differ significantly between NHT-CKD and healthy control cats when compared using unadjusted linear mixed models ([Figure 3](#); [Table 2](#)). Cats with untreated SAH had significantly lower mean concentrations of angiotensin I (geometric mean [95% CI], 9.02 [3.05-26.62] vs 29.78 [21.15-41.94] pmol/L; $P = .008$), angiotensin II (geometric mean [95% CI], 33.63 [12.25-92.26] vs 124.24 [90.52-170.51] pmol/L; $P = .002$), angiotensin III (geometric mean [95% CI], 1.56 [0.88-2.79] vs 7.96 [5.35-11.86] pmol/L; $P = .002$), angiotensin IV (geometric mean [95% CI], 1.43 [0.77-2.66] vs 3.75 [2.59-5.45] pmol/L; $P = .026$), and angiotensin 1-5 (geometric mean [95% CI], 22.59 [7.64-66.8] vs 68.44 [48.45-96.68] pmol/L; $P = .026$) than healthy control cats, and of angiotensin I (geometric mean [95% CI], 9.02 [3.05-26.62] vs 22.89 [15.05-34.81]; $P = .045$) and angiotensin III (geometric mean [95% CI], 1.56 [0.88-2.79] vs 4.69 [2.68-8.20]; $P = .038$) when compared with the NHT-CKD group. Mean angiotensin II concentration was numerically, but not statistically ($P = .05$), higher in NHT-CKD cats when compared to SAH cats. The mean aldosterone concentration was not significantly different between groups. For the surrogate markers of RAAS activity, mean PRA-S was significantly lower and mean AA2 ratio was significantly higher in the SAH group relative to both the NHT-CKD and healthy control groups. There were no significant differences in mean ACE-S or ACE2-S between groups.

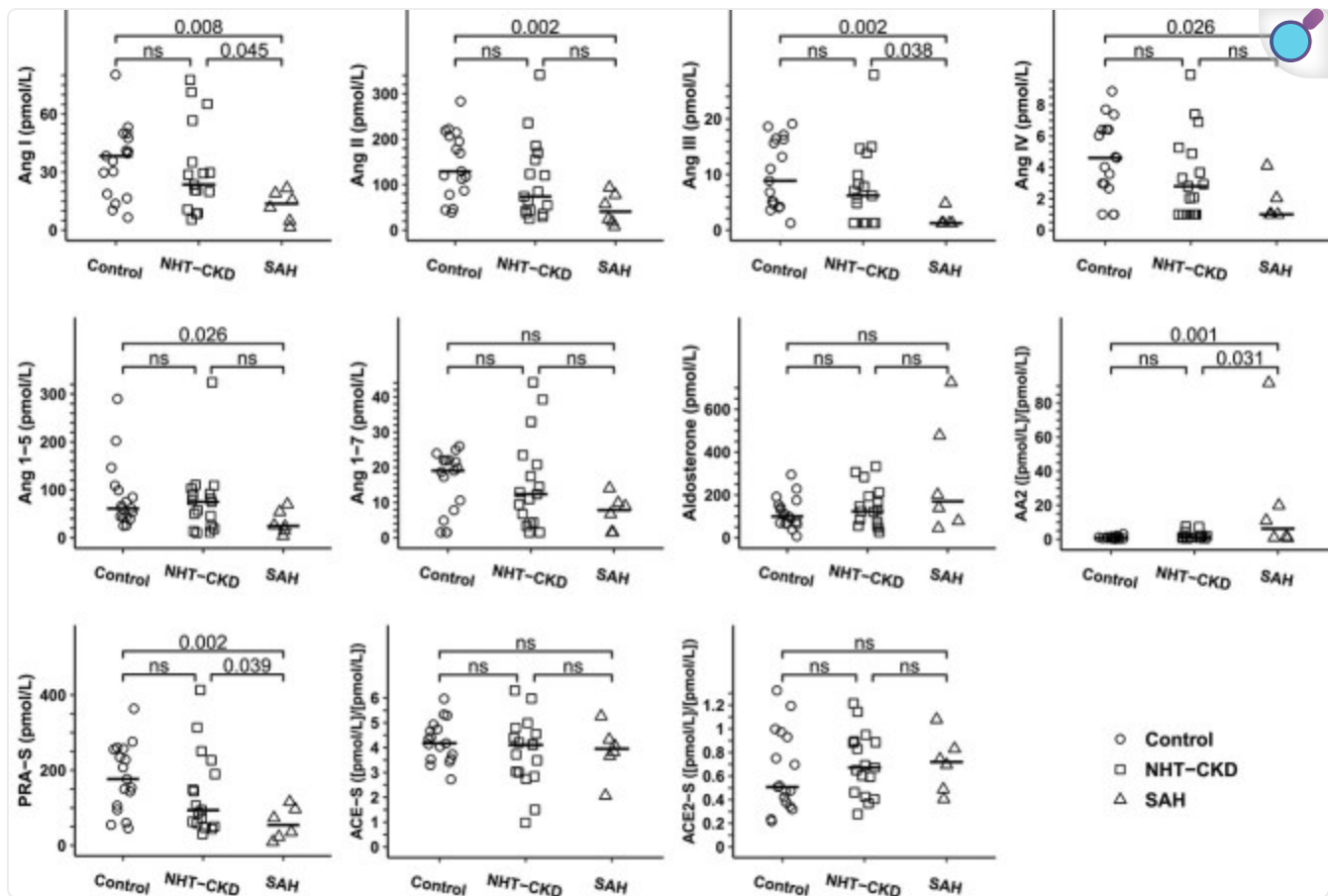
Figure 2.



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Serum equilibrium concentrations of angiotensin peptides and aldosterone in cats with non-hypertensive-CKD (International Renal Interest Society stages 2 or higher), cats with systemic arterial hypertension naïve to antihypertensive therapy (all of which had CKD), and healthy controls that were age- and sex-matched to those in the non-hypertensive-CKD group. The size of each sphere is proportional to median concentrations of each analyte, with median concentrations (pmol/L) shown next to the spheres. Abbreviations: ACE, angiotensin-converting enzyme; ACE2, angiotensin-converting enzyme 2; Aldo, aldosterone; Ang, angiotensin; AP, aminopeptidase; AT1R, angiotensin type-1 receptor; CKD, chronic kidney disease; DAP, aspartyl aminopeptidase; NEP, neutral endopeptidase (neprilysin).

Figure 3.



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Dot plots of circulating renin-angiotensin-aldosterone system markers in cats with NHT-CKD of International Renal Interest Society stages 2 or higher (open squares; $n = 17$), cats with SAH naïve to antihypertensive therapy (open triangles; $n = 6$, all of which had CKD), and healthy controls that were age- and sex-matched to those in the NHT-CKD group ($n = 17$; open circles). Horizontal bars depict the median value for each group. Statistical comparisons between groups were conducted using linear mixed-effect models to account for paired matching between NHT-CKD and healthy control cats. No adjustment for the feeding of a therapeutic renal diet or age was included in this analysis. Significant P -values are noted above relevant brackets. Abbreviations: AA2, angiotensin-to-angiotensin II ratio, a surrogate measure of adrenal responsiveness to angiotensin II; ACE-S, surrogate measure of angiotensin-converting enzyme (ACE) activity; ACE2-S, surrogate measure of ACE2 activity; Ang, angiotensin; ns, not statistically significant; NHT-CKD, non-hypertensive

chronic kidney disease; PRA-S, surrogate measure of plasma renin activity; SAH, systemic arterial hypertension.

Table 2.

Results of intergroup comparisons of circulating renin–angiotensin–aldosterone system markers in cats with NHT-CKD (International Renal Interest Society stages 2 or higher; $n = 17$), cats with untreated SAH ($n = 6$, all of which had CKD), and healthy controls ($n = 17$) using linear mixed models with or without including feeding of a renal diet, age, or both as covariates.

Variable	Group comparison	Unadjusted models			Diet-adjusted models			Age-adju
		Estimate	SE	P-value	Estimate	SE	P-value	Estimate
Ang I	Control vs NHT-CKD	0.30	0.312	.603	0.50	0.364	.366	0.26
	Control vs SAH	1.38	0.432	.008	1.66	0.507	.006	1.27
	NHT-CKD vs SAH	1.07	0.432	.045	1.16	0.437	.032	1.01
Ang II	Control vs NHT-CKD	0.52	0.282	.188	0.83	0.319	.043	0.40
	Control vs SAH	1.53	0.417	.002	1.97	0.476	.001	1.22
	NHT-CKD vs SAH	1.02	0.417	.05	1.15	0.412	.023	0.82
Ang III	Control vs NHT-CKD	0.51	0.245	.123	0.85	0.286	.019	0.33
	Control vs SAH	1.57	0.411	.002	2.05	0.457	.000	1.09
	NHT-CKD vs SAH	1.06	0.411	.038	1.20	0.398	.013	0.76
Ang IV	Control vs NHT-CKD	0.51	0.318	.270	0.61	0.376	.259	0.39

Variable	Group comparison	Unadjusted models			Diet-adjusted models			Age-adju
		Estimate	SE	P-value	Estimate	SE	P-value	Estimate
Ang 1-5	Control vs SAH	1.21	0.443	.026	1.35	0.527	.039	0.89
	NHT-CKD vs SAH	0.70	0.443	.270	0.74	0.455	.251	0.50
	Control vs NHT-CKD	0.30	0.290	.566	0.64	0.330	.154	0.12
	Control vs SAH	1.22	0.445	.026	1.70	0.507	.005	0.73
Ang 1-7	NHT-CKD vs SAH	0.92	0.445	.113	1.06	0.439	.055	0.61
	Control vs NHT-CKD	0.10	0.341	.956	0.34	0.394	.674	−0.02
	Control vs SAH	0.73	0.472	.278	1.07	0.553	.142	0.44
	NHT-CKD vs SAH	0.64	0.472	.378	0.74	0.477	.282	0.45
Aldo	Control vs NHT-CKD	−0.22	0.339	.793	−0.16	0.400	.916	−0.27
	Control vs SAH	−0.79	0.470	.227	−0.70	0.561	.432	−0.91
	NHT-CKD vs SAH	−0.57	0.470	.457	−0.54	0.484	.509	−0.64
	Control vs NHT-CKD	−0.51	0.267	.167	−0.67	0.329	.128	−0.46
AA2	Control vs SAH	−1.60	0.412	.001	−1.83	0.483	.002	−1.48
	NHT-CKD vs SAH	−1.09	0.412	.031	−1.16	0.418	.023	−1.02

Variable	Group comparison	Unadjusted models			Diet-adjusted models			Age-adju
		Estimate	SE	P-value	Estimate	SE	P-value	Estimate
PRA-S	Control vs NHT-CKD	0.46	0.287	.267	0.75	0.327	.081	0.36
	Control vs SAH	1.53	0.418	.002	1.93	0.481	.001	1.26
	NHT-CKD vs SAH	1.07	0.418	.039	1.19	0.416	.019	0.90
ACE-S	Control vs NHT-CKD	0.50	0.342	.34	0.72	0.399	.189	0.31
	Control vs SAH	0.31	0.474	.788	0.64	0.556	.493	−0.19
	NHT-CKD vs SAH	−0.18	0.474	.922	−0.09	0.480	.982	−0.49
ACE2-S	Control vs NHT-CKD	−0.36	0.346	.569	−0.27	0.410	.788	−0.49
	Control vs SAH	−0.42	0.479	.653	−0.30	0.571	.857	−0.79
	NHT-CKD vs SAH	−0.07	0.479	.989	−0.03	0.492	.998	−0.29

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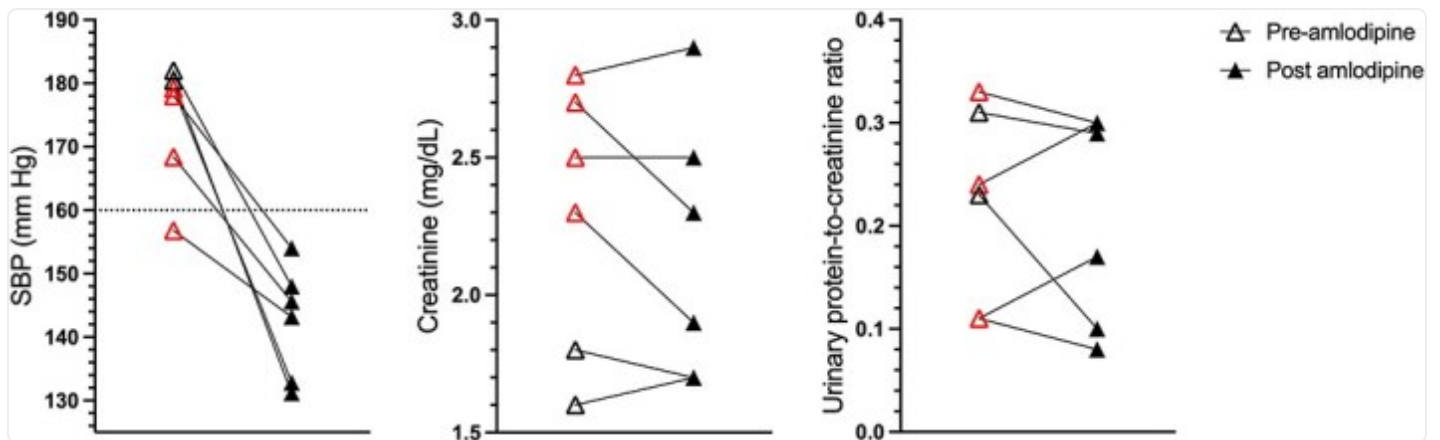
Abbreviations: AA2=angiotensin-to-angiotensin II ratio, a surrogate measure of adrenal responsiveness to angiotensin II; ACE-S, surrogate measure of angiotensin-converting enzyme (ACE) activity; ACE2-S, surrogate measure of ACE2 activity; Aldo=aldosterone; Ang, angiotensin; NHT-CKD=non-hypertensive chronic kidney disease; PRA-S=surrogate measure of plasma renin activity; SAH=systemic arterial hypertension.

Results of intergroup comparisons using linear mixed models adjusted for diet, age, or both are displayed in [Table 2](#). Conclusions for several comparisons differed depending on whether diet- or age-adjustment was conducted. When adjusted for being fed a therapeutic renal diet, without age adjustment, mean concentrations of angiotensin II and III differed significantly between all 3 groups (for both peptides, SAH < NHT-CKD < control). Angiotensins II and III were significantly lower in SAH cats compared with NHT-CKD cats ($P = .023$ and $.013$, respectively) and control cats ($P = .001$ and $P < .001$), and in NHT-CKD cats compared with healthy controls ($P = .043$ and $.019$, respectively). Conclusions for comparisons of all other markers between groups remained unchanged after this adjustment. However, when adjusting for age, or both age and diet, differences between the NHT-CKD and SAH groups in angiotensin I, angiotensin III, AA2, and PRA-S were not statistically significant. In addition, differences between the control and SAH groups for angiotensin IV and 1-5 were also not significant.

Comparison of clinical variables and cRAAS markers in hypertensive cats before and after therapy with amlodipine

All 6 treatment-naïve hypertensive cats were prescribed amlodipine besylate (0.625 mg PO q24h) and reevaluated after 2-4 weeks of therapy. For each cat, follow-up SBP was both lower compared to the pretreatment value and < 160 mmHg, while changes in serum creatinine concentration and UPC were small ([Figure 4](#)). Except for angiotensin IV, median serum equilibrium concentrations of all angiotensin peptides were significantly greater with amlodipine therapy relative to pretreatment ([Figures 5 and 6](#); [Table S4](#)). Concentrations of aldosterone did not differ before and after amlodipine therapy.

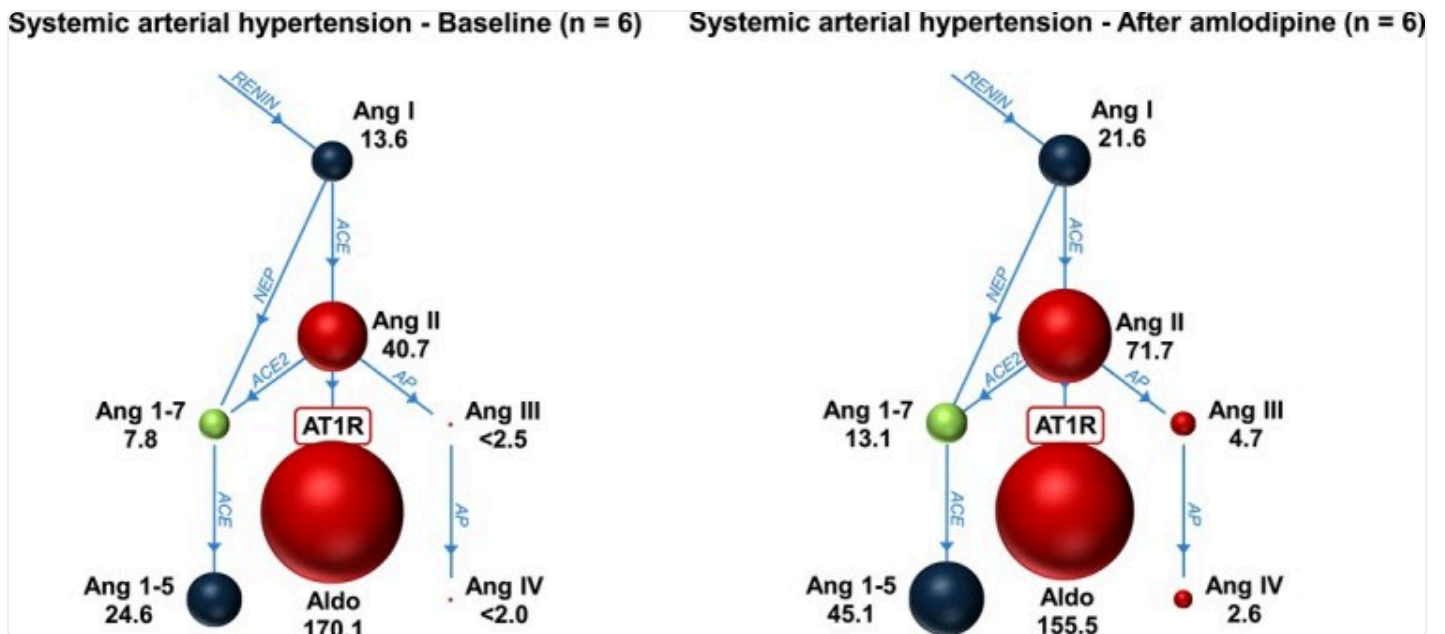
Figure 4.



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Dot plots for systolic blood pressure (SBP), serum creatinine concentration, and urinary protein-to-creatinine ratio in systemically hypertensive cats ($n = 6$, all of which had CKD) before (open triangles) and 2-4 weeks after the start of treatment with amlodipine besylate (0.625 mg PO q24h; solid triangles). Data points from cats with hypertensive ocular damage are denoted in red. Lines connect data from the same cat.

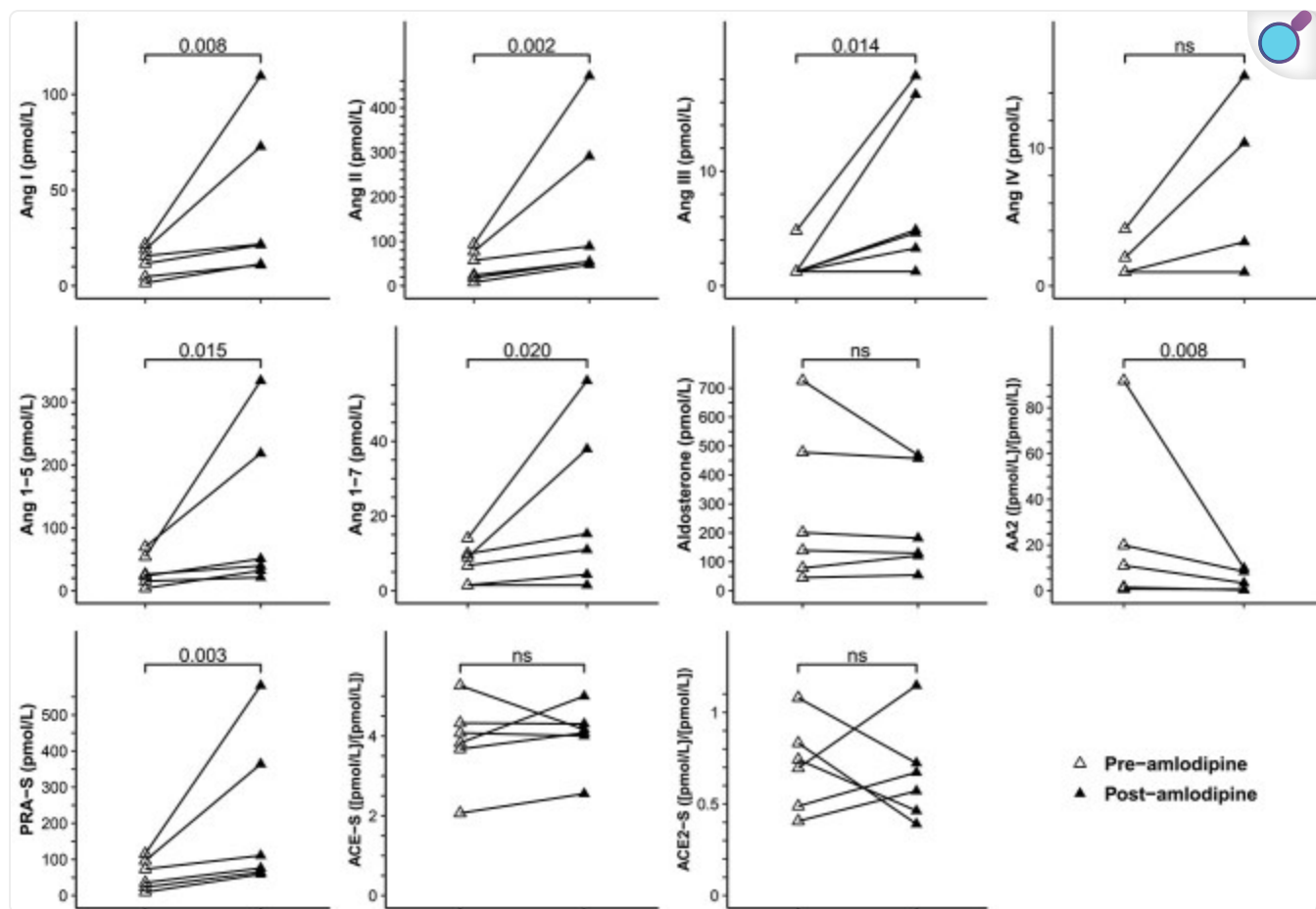
Figure 5.



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Serum equilibrium concentrations of angiotensin peptides and aldosterone of cats with systemic arterial hypertension before (baseline) and 2-4 weeks after the start of treatment with amlodipine besylate (0.625 mg PO q24h). See [Figure 2](#) legend for details and abbreviations.

Figure 6.



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Dot plots for serum equilibrium concentrations of circulating renin-angiotensin-aldosterone system markers in cats with systemic arterial hypertension (all of which had CKD) before (open triangles) and 2-4 weeks after the start of treatment with amlodipine besylate (0.625 mg PO q24h; solid triangles). Statistical comparisons between time points were conducted with the Wilcoxon signed-rank test. Significant *P*-values are noted above relevant brackets. Lines connect data from the same cat. See [Figure 3](#) legend for abbreviations.

Correlation between cRAAS markers and selected clinical, clinicopathologic, and echocardiographic variables

There was significant, though mostly mild to moderate, correlation between several cRAAS markers and SBP, hematocrit, serum sodium concentration, UPC, LA: Ao, and left ventricular FS ([Table 3](#)). Significant negative correlations between SBP and serum equilibrium concentrations of angiotensin I, II, III, IV, and 1-5, and PRA-S were noted, as were positive correlations between SBP and the AA2 ratio. Conversely, hematocrit was positively correlated with serum concentrations of angiotensin I, II, and III, and PRA-S. Serum sodium concentration was positively correlated with serum angiotensin II concentration, whereas UPC was negatively correlated with serum angiotensin 1-5 concentration. Finally, LA: Ao was negatively and LVFS was positively correlated with serum aldosterone concentrations.

Table 3.

Results of Spearman's correlation analysis of circulating renin–angiotensin–aldosterone system markers and select clinical, clinicopathologic, and echocardiographic variables in cats with non-hypertensive chronic kidney disease (International Renal Interest Society stages 2 or higher; $n = 17$), cats with untreated systemic arterial hypertension ($n = 6$; all of which had CKD), and healthy controls ($n = 17$). Significant P -values and corresponding correlation coefficients (ρ) are indicated in bold.

Variable		SBP	Hct	sSodium	sPotassium	sCreatinine	sSDMA	UPC
Ang I	P -value	.009	.038	.085	.78	.24	.07	.24
	ρ	−0.42	0.33	0.28	0.05	−0.19	−0.29	−0.19
Ang II	P -value	.005	.038	.33	.66	.15	.12	.15
	ρ	−0.44	0.33	0.16	0.07	−0.23	−0.25	−0.23
Ang III	P -value	.006	.012	.51	.89	.14	.10	.18
	ρ	−0.44	0.40	0.11	0.02	−0.24	−0.26	−0.22
Ang IV	P -value	.012	.055	.22	.46	.10	.11	.30
	ρ	−0.40	0.31	0.20	0.12	−0.26	−0.26	−0.17
Ang 1-5	P -value	.011	.25	.54	.85	.67	.39	.039
	ρ	−0.40	0.19	0.10	0.03	−0.07	−0.14	−0.33
Ang 1-7	P -value	.051	.066	.039	.89	.72	.19	.36
	ρ	−0.31	0.29	0.33	0.02	−0.06	−0.21	−0.15
Aldo	P -value	.61	.58	.14	.73	.87	.99	.80
	ρ	0.09	0.09	0.24	−0.06	0.03	0.00	0.04
AA2	P -value	.035	.31	.56	.73	.19	.27	.47
	ρ	0.34	−0.16	0.10	−0.06	0.21	0.18	0.12
PRA-S	P -value	.006	.04	.23	.76	.16	.11	.15
	ρ	−0.43	0.33	0.19	0.05	−0.23	−0.26	−0.23
ACE-S	P -value	.41	.51	.25	.92	.59	.96	.63
	ρ	−0.14	0.11	−0.19	−0.02	−0.09	−0.01	−0.08

Variable		SBP	Hct	sSodium	sPotassium	sCreatinine	sSDMA	UPC
ACE2-S	<i>P</i> -value	.52	.36	.45	.51	.27	.84	.11
	ρ	-0.11	-0.15	-0.12	-0.11	0.18	0.03	-0.25

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Abbreviations: AA2, angiotensin-to-angiotensin II ratio, a surrogate measure of adrenal responsiveness to angiotensin II; ACE-S, surrogate measure of angiotensin-converting enzyme (ACE) activity; ACE2-S, surrogate measure of ACE2 activity; Aldo, aldosterone; Ang, angiotensin; Hct, hematocrit; LA:Ao, left-ventricular-to-aorta ratio; LVFS, left ventricular fractional shortening; maxLVWTd, maximum left ventricular wall thickness in diastole; PRA-S, surrogate measure of plasma renin activity; sCreatinine, serum creatinine concentration; sPotassium, serum potassium concentration; sSDMA, serum symmetric dimethylarginine concentration; UPC, urinary protein-to-creatinine ratio.

Discussion

In the present study, serum equilibrium concentrations of angiotensin peptides and aldosterone were compared between cats with NHT-CKD, untreated SAH (in all cases with concurrent CKD as the presumed cause), and apparently healthy cats of similar age and sex relative to those of the NHT-CKD group. Evidence of cRAAS activation in cats with hypertensive or non-hypertensive CKD was lacking. Regardless of whether unadjusted or diet-, age-, or both diet- and age-adjusted models were used for intergroup comparisons, mean concentrations of angiotensin I, II, and III were consistently lower in untreated hypertensive cats relative to healthy cats, as was the value for the surrogate marker of PRA (PRA-S) used. Although the overall pattern indicated that cRAAS activity was lower in SAH cats, intermediate in NHT-CKD cats, and higher in healthy cats, controlling for diet, age, or both altered whether differences in various cRAAS markers between groups were statistically significant. For example, when adjusted for the feeding of a therapeutic renal diet but not age, mean angiotensin II and III concentrations were significantly different between cats with treatment-naïve hypertensive CKD, cats with NHT-CKD, and healthy cats (listed from lowest to highest mean concentrations). Although serum aldosterone concentrations did not differ between groups, the AA2 ratio was higher in the SAH group relative to healthy control groups across all statistical models, suggesting a circulating angiotensin II-independent release of aldosterone in those cats. Hypertensive cats treated with amlodipine experienced activation of the classical and

alternative cRAAS characterized by a significant increase in circulating concentrations of almost all measured angiotensin peptides (ie, angiotensin I, II, III, 1-7, and 1-5) and PRA-S.

The findings of our study are similar to those of a previous large study of 196 cats aged 9 years or older that were presented to a first-opinion practice in the United Kingdom. In that study, PRA and plasma aldosterone concentrations were compared between cats grouped according to the presence or absence of azotemia (ie, plasma creatinine > 2.0 mg/dL) and systemic BP status.⁸ Directly measured PRA was significantly lower in azotemic and non-azotemic cats with untreated systemic hypertension compared with non-azotemic, normotensive cats. In addition, plasma aldosterone concentrations were higher in azotemic cats with untreated systemic hypertension compared to azotemic or non-azotemic normotensive cats. Although our assay techniques and cutoffs for defining renal functional impairment and systemic hypertension differ from those of the previous study, we similarly documented lower PRA-S in hypertensive cats than in normotensive cats. While no significant differences in serum equilibrium concentrations of aldosterone were observed in the present study, only 6 hypertensive cats were evaluated, and a type II error cannot be excluded.

In the previous, larger study,⁸ plasma aldosterone-to-renin ratio was higher in azotemic hypertensive cats than in azotemic normotensive cats, suggesting that the higher aldosterone observed in hypertensive cats is not driven by an increase in PRA and downstream increase in circulating angiotensin II. Somewhat similarly, the present study found a significantly higher mean AA2 ratio in hypertensive cats when compared with the control group across all statistical models and to the NHT-CKD in models without age adjustment, though this appeared to be largely driven by data from 2 individuals. Others who have evaluated smaller samples of cats ($n \leq 16$) have also documented either the same or lower PRA and higher plasma aldosterone concentrations in cats with hypertensive CKD compared to healthy controls,^{6,7} again suggesting that aldosterone release appears independent of angiotensin II production in these cats, as is true for people with low-renin/high aldosterone essential hypertension.³²

Lower-than-normal PRA in the setting of systemic hypertension is an expected physiologic response and suggests normal regulation of the RAAS; that is, elevations in systemic BP should be sensed by the juxtaglomerular apparatus and suppress renin release.³⁰ As expected from normal physiology, we found that the concentrations of almost all angiotensin peptides and PRA-S were negatively correlated with SBP when all cats were considered. This seemingly contradicts the hypothesis that increased activation of the cRAAS is partially responsible for the development of systemic hypertension in patients with CKD. Interestingly, despite our and others' observation of lower PRA (or its surrogate) and circulating angiotensin II concentrations in untreated

hypertensive cats, selective blockade of the angiotensin II type 1 receptor using telmisartan significantly reduces systemic BP in both hypertensive^{33,34} and normotensive³⁵ cats. In recent decades, the role of the intrarenal renin–angiotensin system (RAS) in the regulation of kidney function and BP and in the pathophysiology of SAH has been increasingly highlighted.^{36,37} Notably, regulation of the intrarenal RAS appears to be independent of that of the circulating system,³⁸ and markers of cRAAS activity are poor predictors of renal angiotensin peptide concentrations in cats.³⁹ Intrarenal RAS activity remains uncharacterized in cats with hypertensive CKD, as reliable information on SBP is lacking in prior studies.⁴⁰⁻⁴² Therefore, it is possible (though unproven) that activation of the intrarenal RAS, uncoupled from and undetectable through measurement of the cRAAS, could contribute to SAH in cats with CKD.

In contrast with the present and other studies of cats with spontaneous CKD,⁶⁻⁸ previous work in surgically induced CKD revealed that cats with systemic hypertension after remnant-wrap have higher PRA and plasma aldosterone concentrations than control cats,⁹ and that cats undergoing 7/8 nephrectomy have significantly increased PRA and plasma concentrations of angiotensin I, angiotensin II, and aldosterone approximately 2 months after surgery, relative to baseline.⁴ It would be reasonable to conclude that these findings are specific to surgical models created by reductions in functional renal mass. However, a recent study that used a remnant-kidney CKD model found no difference in serum equilibrium concentrations of angiotensin peptides and aldosterone using RAS-Fingerprint in young adult, purpose-bred cats with induced CKD ($n = 15$, of which 6 were systemically hypertensive and treated with amlodipine) compared to life-stage- and sex-matched, community-owned healthy control cats ($n = 15$).⁴³ In that study, serum aldosterone concentrations and AA2 ratio were higher in cats with induced CKD, even after adjustment for SBP and amlodipine treatment. The time between kidney injury and evaluation of cRAAS markers could factor importantly into these studies' conclusions, as the 2 demonstrating greater-than-normal PRA in cats with hypertensive CKD^{4,9} evaluated markers within 3 months of injury, whereas the most recent, which did not demonstrate differences in angiotensin peptides, evaluated markers at least 27 months after injury.⁴³ In addition, whereas diet was standardized across groups in the earlier studies,^{4,9} in the most recent,⁴³ cats with induced CKD were fed a therapeutic renal diet while control cats were fed maintenance diets. Dietary intakes of sodium and potassium have a known influence on RAAS activity.^{11,12} Compared to maintenance diets, therapeutic renal diets for cats have lower sodium and higher potassium content,⁴⁴ which would be expected to stimulate the RAAS and adrenal release of aldosterone. This would amplify or mask group differences in circulating concentrations of RAAS markers, depending on the group receiving the diet higher in sodium and potassium. In the present study, only 8/17 (47%) cats with NHT-CKD and 4/6 (67%) cats with SAH were fed a therapeutic renal diet, suggesting that dietary influences of lesser magnitude would be expected here compared to the earlier study. Indeed, statistical conclusions for intergroup comparisons of only 2 RAAS components, serum angiotensin II and III,

were altered when these comparisons were adjusted for diet. However, after this adjustment, but not before, angiotensin II and III concentrations were significantly *lower* in NHT-CKD cats relative to healthy controls, and angiotensin II was significantly lower in SAH cats relative to NHT-CKD cats (though this effect was lost if age was concurrently adjusted for). Because a therapeutic renal diet would be expected to increase RAAS activity, the fact that 2 groups (NHT-CKD and SAH) in which these diets were fed with the highest frequency had lower RAAS activity than the group (healthy controls) for which they were not used (after adjustment for diet) suggests that diet might have contributed somewhat by counteracting or masking the tendency of other factors to suppress RAAS activity.

In the present study and the above-cited study,⁸ which evaluated client-owned cats, PRA or its surrogate significantly increased in animals receiving amlodipine, relative to pretreatment values. Since the design of our study, 3 other published studies reported serum equilibrium concentrations of angiotensin peptides in hypertensive cats treated with amlodipine, showing increased concentrations of components of both the classical and alternative RAAS pathways relative to non-hypertensive cats.^{16,17,43}

Correlation analyses were performed between RAAS markers and various physiologically relevant clinical and clinicopathologic variables. Due to the exploratory nature of this analysis, *P*-values reported were not adjusted for multiple comparisons, and findings require further validation. Several variables were significantly, though weakly ($\rho \leq 0.4$), correlated with cRAAS markers in our study. Significant correlations with $\rho > 0.4$ were only found for various cRAAS markers and each of SBP and LA:Ao. In a previous study,¹⁷ cats with cardiomyopathy had significantly higher serum equilibrium concentrations of angiotensin I, angiotensin 1-7, and aldosterone, and PRA-S values than healthy cats. In our study, which excluded cats with substantial heart disease, serum aldosterone concentrations were significantly negatively correlated with LA:Ao, which could be linked to the effects of volume status on the RAAS. For example, blood volume contracture, which would be accompanied by a smaller left atrial size (and therefore, lower LA:Ao), would be expected to be associated with RAAS activation, although angiotensin II was not significantly correlated with this variable.

Strengths of the present study include the comprehensive screening of cats, which included echocardiography to characterize co-existing cardiac pathologies; the exclusion of cats with concurrent conditions and cats receiving medications known to affect the RAAS; the evaluation of both classical and alternative RAAS pathways; and the recruitment of control cats that were age- and sex-matched to the NHT-CKD group.

This study also has important limitations. First, the sample size was small, particularly in the SAH group, which limits statistical power to detect mild or moderate differences between groups, and nonsignificant findings should be interpreted cautiously. Also, as cats were recruited from those regularly presented to a veterinary teaching hospital, time of day of sample collection was not standardized. Because RAAS activity is subject to circadian periodicity,⁴⁵ this omission could have confounded our results. In addition, diet was not standardized, although all cats were fasted prior to sampling, which should decrease the influence of feeding on the cRAAS.⁴⁶ In addition, neither imaging of the urinary tract nor measurement of blood or urinary markers of active kidney injury was performed during screening, which limited our characterization of the cats' kidney diseases; however, all cats were clinically stable at the time of enrollment. Finally, our assessment of the RAAS was limited to circulating markers of this system, and the intrarenal RAS was not explored.

In conclusion, we did not detect activation of the cRAAS in cats with NHT-CKD or SAH, relative to age- and sex-matched controls. The intrarenal RAS should be explored in cats with these conditions for a more complete understanding of the role of this system in their pathophysiology.

Supplementary Material

aalag079_cRAAS_in_cats_with_CKD_or_SAH_supplement_R1

[aalag079_craas_in_cats_with_ckd_or_sah_supplement_r1.docx](#) (26.8KB, docx)

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Abbreviations

2D

2-dimensional

BP

blood pressure

CKD

chronic kidney disease

cRAAS

circulating renin–angiotensin–aldosterone system

IRIS

International Renal Interest Society

LA:Ao

left-atrial-to-aortic dimension ratio in short axis

FS

left ventricular fractional shortening

maxLVWTd

maximum end-diastolic left ventricular wall thickness

NHT-CKD

non-hypertensive chronic kidney disease

RAAS

renin–angiotensin–aldosterone system

RAS

renin–angiotensin system

SAH

systemic arterial hypertension

SBP

systolic blood pressure

SDMA

symmetric dimethylarginine

TT4

total thyroxine

UPC

urinary protein-to-creatinine ratio

USG

Urine specific gravity

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Conflicts of interest

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Data availability

All data are included in the manuscript or supplemental material.

Off-label antimicrobial declaration

The authors declare no off-label use of antimicrobials.

Institutional animal care and use committee or other approval declaration

Approved by the University of Georgia College of Veterinary Medicine Clinical Research Committee (CR-594).

Human ethics approval declaration

The authors declare human ethics approval was not needed.

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Associated Data

This section collects any data citations, data availability statements, or supplementary materials included in this article.

Supplementary Materials

aalag079_cRAAS_in_cats_with_CKD_or_SAH_supplement_R1

[aalag079 craas in cats with ckd or sah supplement r1.docx](#) (26.8KB, docx)

Data Availability Statement

All data are included in the manuscript or supplemental material.

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